

Hitchin Harmonic Spinors Notes

1 Outline

1. Preliminaries

- (a) Clifford algebra
- (b) spin groups and representation from last time
- (c) lifting connection to spin
- (d) spin c

2. theorem statement and proof

In the last talk, Nazif told us about Clifford algebras, $\text{spin}(k)$, and Dirac Operators associated to Clifford bundles.

In my lecture, I will cover theorem 1.1, the vanishing theorem. We will first introduce the $\text{Spin}^c(k)$ group, then take a closer look at the Lie algebras of $\text{Spin}(k)$ and $\text{SO}(k)$.

2 The Lie algebra $\mathfrak{spin}(k)$

(Some of this section is from Spin Geometry along with Hitchin)

The group of units $Cl^\times(k)$ is a Lie group with Lie algebra $\mathfrak{cl}^\times(k) \simeq (Cl(k), [\cdot, \cdot])$, where $[\cdot, \cdot]$ is the commutator in $Cl(k)$.

Since $Cl(k)$ is in fact a matrix group, the Adjoint action of the Lie group of units is just given by conjugation.

$\text{Spin}(k)$ is a Lie subgroup of $Cl^\times(k)$, so its Lie algebra $\mathfrak{spin}(k)$ is a Lie subalgebra of $\mathfrak{cl}^\times(k) \simeq (Cl(k), [\cdot, \cdot])$, with the same product.

Recall that there are canonical embeddings $\wedge^p \mathbb{R}^n \subset Cl(k)$ for all p . Then the explicit subalgebra is given by $\wedge^2 \mathbb{R}^n$. To see this, note the Lie algebra is spanned by tangent vectors to $\text{Spin}(k)$ at 1. Fix an orthonormal basis $\{e_1, \dots, e_n\}$ of $\mathbb{R}^n$ and for any pair $i < j$ consider the curve

$$\gamma(t) = \cos(t) + \sin(t)e_i e_j$$

whose velocity at $\gamma(0) = 1$ is $e_i e_j$. This curve lies in $\text{Spin}(k)$ by definition. Then $\mathfrak{spin}(k)$ contains the vector subspace $\text{span}_{\mathbb{R}}\{e_i e_j\}_{i < j} = \wedge^2 \mathbb{R}^n$. Since $\dim_{\mathbb{R}}(\mathfrak{spin}(k)) = n(n-1)/2$, they are equal.

Now consider $\mathfrak{so}(k)$, consisting of skew-symmetric $k \times k$ real valued matrices. We can identify this as a vector space with $\wedge^2 \mathbb{R}^n = \text{span}_{\mathbb{R}}\{e_i \wedge e_j : i < j\}$ by sending a basis element in $\mathfrak{so}(k)$ with (i, j) entry equal to 1 to the element $e_i \wedge e_j$, with $e_i \wedge e_j(x) = \langle e_i, x \rangle e_j - \langle e_j, x \rangle e_i$.

Last lecture we introduced the double cover $\text{Spin}(k) \rightarrow \text{SO}(k)$ with kernel $\{-1, +1\}$. This is in fact a surjective homomorphism of Lie groups, and induces an isomorphism of Lie algebras $\Phi : \mathfrak{spin}(k) \rightarrow \mathfrak{so}(k)$.

Proposition. *The Lie algebra isomorphism Φ induced by the double cover $\text{Spin}(k) \rightarrow \text{SO}(k)$ is given on basis elements by*

$$\Phi(e_i e_j) = 2e_i \wedge e_j,$$

under the above identification of $\mathfrak{so}(k)$ with $\wedge^2 \mathbb{R}^n$.

To prove this, consider again $\gamma(t) = \cos(t) + \sin(t)e_i e_j$ in $\text{Spin}(k)$ with $\gamma(0) = 1$ and $\gamma'(0) = e_i e_j$.

The cover here is actually induced by the adjoint action of Spin on U , which maps into $\text{SO}(k)$ with kernel 1 and -1 . Then the image of $\gamma(t)$ applied to a vector x is $\gamma(t)x\gamma(t)^{-1}$, so that Φ , the differential of this map, is the commutator $\Phi(\gamma(t))(x) = [\gamma(t)', x]$.

Relative to a correct choice of basis, if the connection form is given locally by ω_{ij} , then the corresponding connection form of the lifted connection on the spinor bundle is given by

$$\frac{1}{4} \sum_{i,j} \omega_{ij} e_i e_j \in \Gamma(\text{End}(V) \otimes T^*)$$

3 The complex spin group $\text{Spin}^c(k)$

We briefly define the group $\text{Spin}^c(k)$. Let $\text{Spin}^c(k) = \text{Spin}(k) \times_{\mathbb{Z}_2} S^1$ in $Cl(k) \otimes \mathbb{C}$. Then we have the following short exact sequences

$$1 \rightarrow S^1 \rightarrow \text{Spin}^c \rightarrow SO \rightarrow 1$$

$$1 \rightarrow \text{Spin} \rightarrow \text{Spin}^c \rightarrow S^1 \rightarrow 1,$$

thus giving a map

$$\text{Spin}^c \rightarrow SO \times S^1$$

with kernel $S^1 \cap \text{Spin} = \{-1, +1\}$, showing that Spin^c is a double cover of $SO \times S^1$. We say a manifold is “ Spin^c ” if its frame bundle lifts to a principal Spin^c -bundle via the first sequence. The second gives an associated S^1 bundle L on X for any Spin^c structure.

To get a connection on the principal Spin^c bundle, we must choose a hermitian connection on L

4 The Vanishing Theorem

To start, let X be a Spin^c manifold, and let $i\theta$ be the curvature form of the associated line bundle L . Then $\theta \in \Gamma(\lambda^2 T^*)$ defines, via the Riemannian metric, a skew-symmetric endomorphism of T . Suppose its eigenvalues are $\pm\lambda_1, \dots, \pm\lambda_m$, where $\dim X = 2m$ or $2m + 1$.

Theorem. *Let X be a Spin^c manifold with scalar curvature $R \geq 4 \sum |\lambda_i|$ and strict inequality at some point. Then X admits no harmonic spinors.*

Proof: Recall the Dirac operator P given by the composition

$$\Gamma(V) \xrightarrow{D} \Gamma(V \otimes T^*) \xrightarrow{m} \Gamma(V),$$

where D is the connection and m is Clifford multiplication, after identifying $T^* \simeq T$ using the metric. Since D and m commute, we have $P^2\varphi = m^2 D^2\varphi$, where $m^2 : \Gamma(V \otimes T^* \otimes T^*) \rightarrow \Gamma(V)$ is defined by $m^2(\varphi \otimes \alpha \otimes \beta) = \alpha \cdot \beta \cdot \varphi$. Under the identification of the Clifford algebra with the exterior algebra, the multiplication becomes

$$\alpha \cdot \beta = \alpha \wedge \beta - (\alpha, \beta).$$

Then we have

$$P^2\varphi = m^2 D^2\varphi = \sum_{i,j} m^2(\nabla_i \nabla_j \varphi \otimes e_i \otimes e_j) = \sum_{i,j} (e_i \wedge e_j - (e_i, e_j)) \nabla_i \nabla_j \varphi = \quad (1)$$

$$\sum_{i < j} e_i \cdot e_j (\nabla_i \nabla_j - \nabla_j \nabla_i) \varphi - \sum_i \nabla_i \nabla_i \varphi = \Omega \cdot \varphi - \text{tr} D^2 \varphi, \quad (2)$$

where $\Omega \cdot$ is the “Clifford contracted” Riemann curvature tensor.

Now, since

$$D(D\varphi, \varphi) = (D^2\varphi, \varphi) + (D\varphi, D\varphi) \in \Gamma(T^* \otimes T^*),$$

we have

$$(\text{tr} D^2 \varphi, \varphi) = -\langle D\varphi, D\varphi \rangle + d^*(D\varphi, \varphi),$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product on $V \otimes T^*$.

Then $(P^2 \varphi, \varphi) = (\Omega \cdot \varphi, \varphi) + \langle D\varphi, D\varphi \rangle - d^*(D\varphi, \varphi)$, and integrating over X we obtain

$$\int_X (P^2 \varphi, \varphi) dV = \int_X (\Omega \cdot \varphi, \varphi) + \langle D\varphi, D\varphi \rangle dV$$

This tells us more about the kernel of P^2 . Suppose $\Omega \geq 0$ and $P^2 \varphi = 0$. Then the positivity of $\langle D\varphi, D\varphi \rangle$ implies $D\varphi = 0$.

Moreover, if $\Omega > 0$ at some point in X , we must have $\varphi = 0$ at that point, otherwise $(\Omega \cdot \varphi, \varphi)$ would be positive. Then since $D\varphi = 0$, this implies $\varphi = 0$ everywhere.

Now we will examine Ω in more detail. We have $\Omega \in \Gamma(L(\text{Spin}^c) \otimes \wedge^2 T^*)$, where $L(\text{Spin}^c)$ is the bundle of Lie algebras associated to the principal Spin^c bundle by the adjoint representation.

We have $L(\text{Spin}^c) = L(SO) \oplus \mathbb{R}$, where $\mathbb{R}$ is acted on trivially by the adjoint representation. Hence, we can decompose $\Omega = \Omega_1 + \Omega_2$, where $\Omega_1 \in \Gamma(L(P_{SO}) \otimes \wedge^2 T^*)$ and $\Omega_2 \in \Gamma(\wedge^2 T^*)$.

1. For the Ω_2 component, we need to show its eigenvalues are $\sum \pm \lambda_k$. We write

$$\Omega_2 = i \sum \lambda_k e_{2k-1} e_{2k}.$$

The operators $E_k = e_{2k-1} e_{2k}$ decompose V into 2^m subspaces $\varphi(k_1, \dots, k_m)$, $k_i = \pm 1$, such that on each $\varphi(k_1, \dots, k_m)$, E_j acts as $i k_j$. Then the eigenvalues of Ω_2 are $\pm \lambda_1 \pm \dots \pm \lambda_m$, and the smallest is $-\sum |\lambda_i|$.

2. It is a standard calculation that $\Omega_1 = \frac{1}{4}R$, with R the scalar curvature. Then Ω is positive whenever $\frac{1}{4}R \geq \sum |\lambda_i|$